ITA06020 MACHINE LEARNING

Experiment 16 – Comparison of Classification Algorithms

Aim:
To compare the performance of different classification algorithms using the Iris dataset.

Procedure:

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing data.
4. Apply KNN, Decision Tree, Naive Bayes and Logistic Regression algorithms.
5. Calculate and compare their accuracy.

Dataset: Iris Flower Dataset (built-in Scikit-learn dataset).

Python Code:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.pipeline import make_pipeline

from sklearn.neighbors import KNeighborsClassifier

from sklearn.tree import DecisionTreeClassifier

from sklearn.naive_bayes import GaussianNB

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42, stratify=y

)
```

```
models = {
    "KNN": make_pipeline(StandardScaler(), KNeighborsClassifier(n_neighbors=5)),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Naive Bayes": GaussianNB(),
    "Logistic Regression": make_pipeline(
        StandardScaler(), LogisticRegression(max_iter=1000)
    )
}
for name, model in models.items():
    model.fit(X_train, y_train)
    predictions = model.predict(X_test)
    accuracy = accuracy_score(y_test, predictions)
    print(name, "Accuracy:", round(accuracy * 100, 2), "%")
```

Output:

```
KNN Accuracy: 93.33 %
Decision Tree Accuracy: 93.33 %
Naive Bayes Accuracy: 96.67 %
Logistic Regression Accuracy: 93.33 %
```

Result:
The classification algorithms were successfully implemented and compared. Naive Bayes achieved the highest accuracy of 96.67% on the test dataset.